

Queues

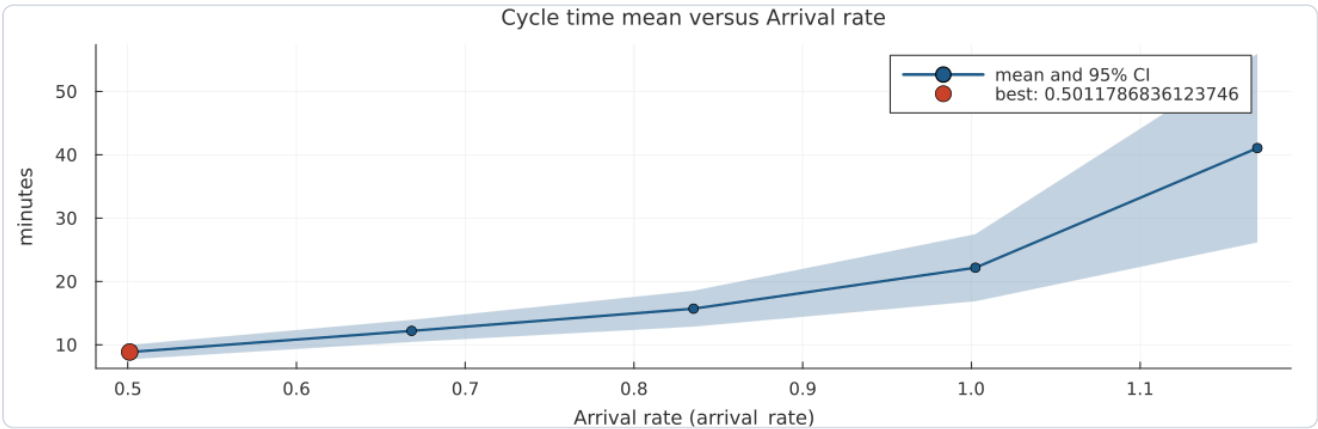
Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
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Queues and capacity

The first thing to check in a queueing model is that it agrees with Erlang, and the second is what happens when the load moves.

Widening the bottleneck



Sweep

Arrival rate

Value	Throughput mean	Wait mean	Utilisation mean	Cycle time mean
0.501	0.49	—	0.343	—
0.668	0.645	—	0.454	—
0.835	0.806	—	0.565	—
1.002	0.963	—	0.676	—
1.169	1.103	—	0.773	—

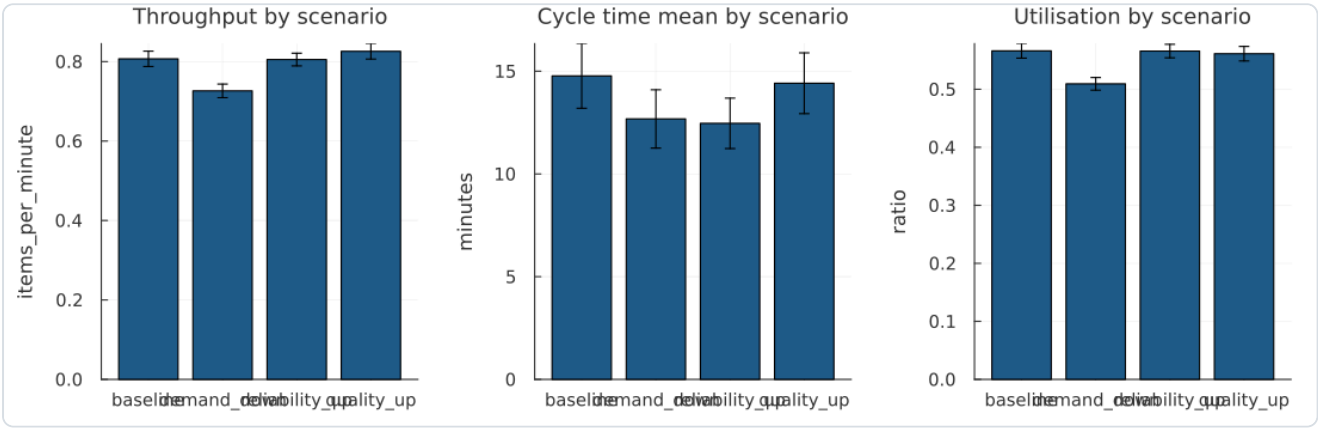
The best arrival_rate by cycle_time_mean is 0.5011786836123746 (mean 8.862 ± 1.135).

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.807	0.726	-0.081	0.00305	-9.99	worse
demand_down	cycle_time_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	wait_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	utilisation	0.566	0.509	-0.057	0.00231	-10	worse

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	completed	2905	2614	-290.2	11	-9.992	worse
reliability_up	throughput	0.807	0.805	-0.0017	0.00397	-0.211	indistinguishable
reliability_up	cycle_time_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	wait_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	utilisation	0.566	0.566	-0.00036	0.00104	-0.064	indistinguishable
reliability_up	completed	2905	2898	-6.375	14.5	-0.219	indistinguishable
quality_up	throughput	0.807	0.826	0.019	0.00223	2.351	better
quality_up	cycle_time_mean	14.8	14.4	-0.357	0.157	-2.414	better
quality_up	wait_mean	14.8	14.4	-0.357	0.157	-2.414	better
quality_up	utilisation	0.566	0.561	-0.00489	0.000593	-0.863	worse
quality_up	completed	2905	2973	68.2	7.941	2.35	better



Scenarios

Best by cycle_time_mean: **reliability_up** (compared with baseline, common random numbers on every replication).